



TRIOS CYBER

LABORATORY PRACTICE 30

Final Practical SOC Investigation

Name: Matthew Josue Portilla Morejon

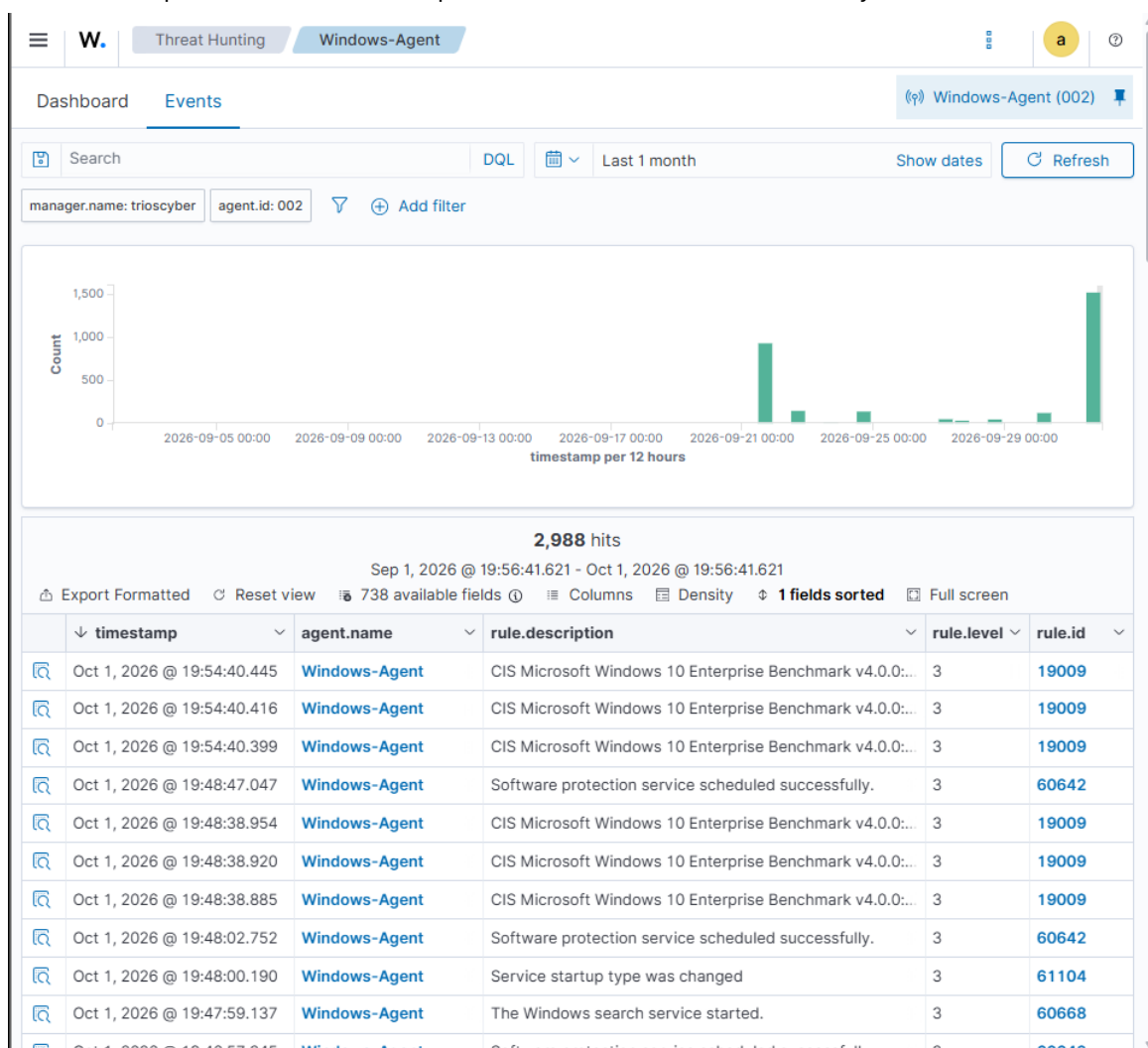
1. Investigation Objective and Scenario

This final practical investigation followed one safe, self-generated Windows service-creation scenario from telemetry collection through alert identification, triage, log investigation, evidence review, MITRE ATT&CK mapping, cleanup, and final disposition. The controlled service TriosLab30 was created on the monitored Windows endpoint and then investigated in both Windows System logs and Wazuh.

Evidence item	Observed value	Purpose / result
Monitored endpoint	Windows-Agent / ID 002 / 10.10.20.30	Endpoint selected for the investigation
Controlled behavior	Creation of service TriosLab30	Safe event used to generate telemetry
Windows event	System Event ID 7045	Service-installation evidence
Wazuh alert	Rule 61138 / Level 5	New Windows Service Created
MITRE ATT&CK	T1543.003 - Windows Service	Persistence / Privilege Escalation mapping

2. Telemetry Collection and Readiness

Before generating the scenario, Wazuh Threat Hunting showed Windows-Agent (ID 002) actively reporting telemetry. This established that the endpoint-to-Wazuh collection path was available before the test activity was introduced.



Wazuh Threat Hunting showing active telemetry from Windows-Agent (ID 002).

3. Controlled Service Creation

The service TriosLab30 was created from the Windows command line with a harmless command path, demand start, and LocalSystem as the service account. The Service Control command returned CreateService CORRECTO, confirming that Windows accepted the controlled service creation.

```
matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe create TriosLab30 binPath= "
C:\Windows\System32\cmd.exe /c exit" start= demand obj= LocalSystem
[SC] CreateService CORRECTO
```

Controlled creation of the TriosLab30 Windows service.

4. Windows Log Investigation

The Windows System log was queried for Service Control Manager Event ID 7045. The returned record confirms that TriosLab30 was installed on DESKTOP-VOJCDSO. The event preserves the service name, image path C:\Windows\System32\cmd.exe /c exit, user-mode service type, demand start, LocalSystem account, and the event time 2026-10-01T19:58:33.3700000Z.

```
matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe create TriosLab30 binPath= "
C:\Windows\System32\cmd.exe /c exit" start= demand obj= LocalSystem
[SC] CreateService CORRECTO

matthew@DESKTOP-VOJCDSO C:\Users\matthew>wevtutil qe System /q:"*[System[(EventID=7045)]]" /c:5 /rd:true /f:text
Event[0]:
  Log Name: System
  Source: Service Control Manager
  Date: 2026-10-01T19:58:33.3700000Z
  Event ID: 7045
  Task: N/A
  Level: Información
  Opcode: N/A
  Keyword: Clásico
  User: S-1-5-21-3039163649-2597512884-193828224-1001
  User Name: DESKTOP-VOJCDSO\matthew
  Computer: DESKTOP-VOJCDSO
  Description:
Se instaló un servicio en el sistema.

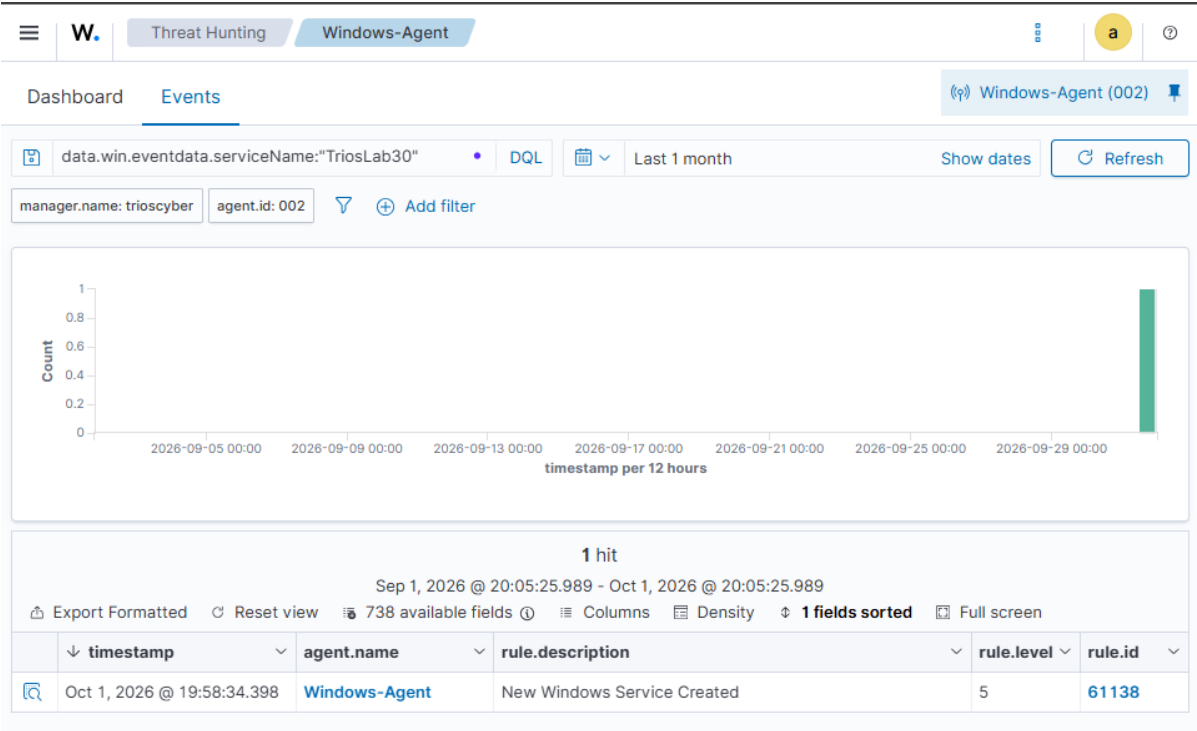
Nombre del servicio: TriosLab30
Nombre del archivo del servicio: C:\Windows\System32\cmd.exe /c exit
Tipo de servicio: servicio de modo usuario
Tipo de inicio de servicio: inicio por solicitud
Cuenta de servicio: LocalSystem
```

Windows System Event ID 7045 confirming installation of TriosLab30.

Field	Observed value	Interpretation
Provider	Service Control Manager	Native Windows source for the service event
Event ID	7045	Service installed in the system
Service name	TriosLab30	Controlled service under investigation
Image path	C:\Windows\System32\cmd.exe /c exit	Harmless command used by the lab service
Service account	LocalSystem	Account recorded for the service
Start type	Demand start	Service configured for on-demand start

5. Wazuh Alert Identification

After the Windows event was generated, Wazuh Threat Hunting was filtered for `data.win.eventdata.serviceName:"TriosLab30"`. One matching alert was returned from Windows-Agent with the description New Windows Service Created, rule ID 61138, and rule level 5.



Wazuh alert generated for the TriosLab30 service-creation event.

Field	Observed value	Meaning
Agent	Windows-Agent / ID 002	Monitored endpoint that generated the alert
Service	TriosLab30	Service matched by the Wazuh search
Rule	61138 / Level 5	New Windows Service Created
Alert time	Oct 1, 2026 @ 19:58:34.398	Central Wazuh event timestamp

6. Detailed Wazuh Evidence and Initial Triage

Wazuh Document Details correlates the Windows event with the monitored endpoint. The record identifies agent ID 002 at 10.10.20.30, Windows-Agent, TriosLab30, the same command image path, LocalSystem, demand start, System channel, Event ID 7045, and Service Control Manager. These fields confirm that the centralized alert matches the native Windows telemetry.

[Table](#) [JSON](#)

<code>_index</code>	wazuh-alerts-4.x-2026.10.02
<code>agent.id</code>	002
<code>agent.ip</code>	10.10.20.30
<code>agent.name</code>	Windows-Agent
<code>data.win.eventdata.accountName</code>	LocalSystem
<code>data.win.eventdata.imagePath</code>	C:\\Windows\\System32\\cmd.exe /c exit
<code>data.win.eventdata.serviceName</code>	TriosLab30
<code>data.win.eventdata.serviceType</code>	servicio de modo usuario
<code>data.win.eventdata.startType</code>	inicio por solicitud
<code>data.win.system.channel</code>	System
<code>data.win.system.computer</code>	DESKTOP-V0JCDS0
<code>data.win.system.eventID</code>	7045
<code>data.win.system.eventRecordID</code>	1784
<code>data.win.system.eventSourceName</code>	Service Control Manager
<code>data.win.system.keywords</code>	0x0000000000000000
<code>data.win.system.level</code>	4
<code>data.win.system.message</code>	"Se instaló un servicio en el sistema. Nombre del servicio: TriosLab30 Nombre del archivo del servicio: C:\\Windows\\System32\\cmd.exe /c exit Tipo de servicio: servicio de modo usuario Tipo de inicio de servicio: inicio por solicitud Cuenta de servicio: LocalSystem"

Detailed Wazuh fields for the endpoint and Windows service event.

Triage question	Observed evidence	Assessment
What happened?	A new Windows service was installed	Service creation requires validation
Which asset?	Windows-Agent / ID 002 / 10.10.20.30	Affected endpoint identified
Which service?	TriosLab30	Controlled service identified
Execution context	cmd.exe /c exit under LocalSystem	Privileged service context recorded
Alert classification	Rule 61138 / Level 5	Wazuh classified the service-creation event
Was the activity authorized?	Yes - self-generated safe laboratory scenario	No evidence of an unauthorized compromise in this test

Analyst assessment: The alert is a valid detection of a real Windows service-creation event. Its security relevance depends on context. In this investigation the activity was intentionally generated for the laboratory, so the event is treated as authorized test activity rather than a confirmed compromise.

7. MITRE ATT&CK Mapping

The lower Wazuh event fields preserve the ATT&CK context assigned to the alert. Wazuh maps the event to T1543.003 - Windows Service and associates it with the Persistence and Privilege Escalation tactics. The mapping is supported by the observed service installation, Service Control Manager provider, LocalSystem account, and the service configuration recorded in the event.

data.win.system.opcode	0
data.win.system.processID	592
data.win.system.providerGuid	{555908d1-a6d7-4695-8e1e-26931d2012f4}
data.win.system.providerName	Service Control Manager
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-10-02T00:58:33.3708996Z
data.win.system.task	0
data.win.system.threadID	1896
data.win.system.version	0
decoder.name	windows_eventchannel
id	1790902714.2933983
input.type	log
location	EventChannel
manager.name	trioscyber
rule.description	New Windows Service Created
# rule.firedtimes	1
rule.groups	windows, windows_system
rule.id	61138
# rule.level	5
rule.mail	false
rule.mitre.id	T1543.003
rule.mitre.tactic	Persistence, Privilege Escalation
rule.mitre.technique	Windows Service
timestamp	Oct 1, 2026 @ 19:58:34.398

Wazuh rule and MITRE ATT&CK fields for the selected alert.

Field	Observed value	Meaning
Rule ID / level	61138 / Level 5	New Windows service alert classification
Technique	T1543.003 - Windows Service	ATT&CK technique assigned by Wazuh
Tactics	Persistence, Privilege Escalation	ATT&CK tactical context
Rule groups	windows, windows_system	Windows system-event classification
Timestamp	Oct 1, 2026 @ 19:58:34.398	Central Wazuh alert time

8. Evidence and Actions Summary

Investigation step	Evidence / result	Status
Collect telemetry	Windows-Agent actively reporting to Wazuh	Verified
Generate safe scenario	TriosLab30 created successfully	Completed
Investigate Windows logs	System Event ID 7045 reviewed	Verified
Identify Wazuh alert	Rule 61138 / Level 5 / New Windows Service Created	Verified
Review detailed evidence	Endpoint, service, account, path, provider, timestamp correlated	Completed
Map ATT&CK behavior	T1543.003 - Windows Service / Persistence / Privilege Escalation	Mapped
Cleanup	TriosLab30 deleted after investigation	Completed

9. Closure and Escalation Recommendation

Closure decision: Close as benign / authorized laboratory activity. The service creation was deliberately generated for the practical investigation, the observed Windows and Wazuh telemetry are consistent with that scenario, and no additional evidence in the collected material indicates an unauthorized compromise.

For an equivalent unexplained event on a production endpoint, the recommended action would be to escalate for validation of the service owner, service name, executable path, service account, start type, initiating change, and surrounding endpoint activity. An unauthorized or suspicious service should remain open for deeper investigation and appropriate containment according to organizational procedures.

10. Laboratory Cleanup

After the evidence was collected and the investigation was completed, the TriosLab30 service was removed from the Windows endpoint. The Service Control command returned DeleteService CORRECTO, confirming cleanup of the test artifact.

```
matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe delete TriosLab30
[SC] DeleteService CORRECTO
```

Removal of the TriosLab30 service after evidence collection.

11. Validation Summary

Requirement	Observed result	Status
Collect telemetry	Windows-Agent visible and reporting in Wazuh	Verified
Identify alert / pattern	New Windows Service Created - rule 61138 / level 5	Verified
Triage	Endpoint, service, execution context, and authorization reviewed	Completed
Investigate logs	Windows System Event ID 7045 examined	Verified
Collect evidence	Windows log + Wazuh alert + Document Details preserved	Completed
ATT&CK mapping	T1543.003 - Windows Service; Persistence / Privilege Escalation	Mapped
Closure recommendation	Authorized lab activity closed; unexplained production case should escalate	Completed

12. Conclusions

The end-to-end SOC investigation was completed successfully using a safe Windows service-creation scenario. Windows recorded the installation of TriosLab30 under System Event ID 7045, and Wazuh correlated the same activity as New Windows Service Created under rule 61138 at level 5. Detailed evidence confirmed the reporting endpoint, service configuration, LocalSystem context, Service Control Manager source, and timestamp. Wazuh mapped the behavior to MITRE ATT&CK T1543.003 - Windows Service under the Persistence and Privilege Escalation tactics. Because the activity was intentionally self-generated for the laboratory and the test service was removed after evidence collection, the case can be closed as authorized activity. The same behavior, if unexplained on a production system, should be escalated for further investigation.